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Networking:

This MESH ALGORITHM Allows OTA UPGRADES

Mesh networks are getting popular and Maven Systems has a unique way of implementing these. Anand Bhandari, head - IoT/Telematics/M2M, Maven Systems, an Internet of Things and M2M product and services company, speaks with Priya Ravindran from EYF

Q. Could you explain the thought process behind the development of your smart mesh network?

A. Smart mesh is one of our flagship designs. It eliminates the need for repeaters and ensures short-range wireless communication. Consider a pipeline project that spans about 200 kilometres. Every kilometre or so there is a pressure detector that detects pressure at the point and boosts it up, if there is a drop, so that whatever is travelling through the pipe, say oil, reaches its destination.

As the pipeline travels through all possible terrains, it is impossible to have a GSM network connectivity throughout. So the easiest way is to have a satellite communication device that would communicate with the server.

Although this is technically viable, the cost and licence regulations offer complications. So we set up a series of RF modules that can calculate pressure and also communicate the same up to a distance of six kilometres, over the entire range of the pipeline. This is not just effective but also economical. It reduces both one-time hardware and month-over-month occurring expenses.

Q. How does your smart mesh network technology work?

A. The modules communicate the information from one to the other, creating a chain, and data hops from one module to the other, until it reaches a gateway that is GPRS-

enabled, commonly called the mobile gateway. This gateway has two interfaces: one for the short-range wireless network and one for the mobile network.

Data can then be transferred to the Cloud and processed according to the end application. This solution becomes perfectly viable from an economical point of view as it is a one-time hardware installation, and we need only one gateway for about 50 metres, cutting down on the recurring cost.

Q. How do you expect this technology to cope with any changes?

A. The concept can be extended to streetlights to achieve lighting automation for a smartcity solution. This mesh algorithm allows OTA upgrades, as once deployed, the system cannot be made to stop. Whenever an enhancement or a bug fix is required, all you have to do is send a command over the Cloud to the mobile gateway. It automatically upgrades all 500 nodes in its vicinity. The whole process can be performed sitting at one place, and within a matter of an hour.

Q. Can the same be implemented with Bluetooth-mesh combination?

A. Mesh networks with Bluetooth will not help our application much. Bluetooth ideally works on 2.4GHz. The inherent problem with high frequencies is that these give a lower range. Our RF modules are based on 865MHz to 868MHz. Thus, their

penetration is about six kilometres, within the specified power range.

Q. What applications would benefit from a BLE-enabled mesh network?

A. Bluetooth Low Energy (BLE) aims at sending smaller packets of data over a maximum distance of about ten metres, targeting different applications, of course. We are working on a product combining BLE and RF for the safety of kids in a school bus. Every child in the bus will have a BLE tag, or an iBeacon, and every bus an iBeacon-to-GPRS gateway.

Every time a child gets on the bus, the driver will get a notification. Every area in the school will have a BLE-to-RF gateway. BLE can send data to the short-range RF, which then connects to a mobile network and data is finally sent to the Cloud.

Q. Would the life of the sensors deployed in far-off areas pose any challenges?

A. We use battery-operated wireless modules that have a power-management algorithm implemented in these, so that sensors can work for several years.

These have two parts. The first is called on-change mechanism; sensors have the intelligence to send data only when there is a change.

The second is called wake-up on RF, where the system has a real-time clock, and the inbuilt algorithm wakes up the RF module at a pre-defined time. **EYF**